

Job Posting: 470351 - Position: Software Development Research Assistant

Co-op Work Term Posted:	2026 - Fall
App Deadline	05/19/2026 09:00 AM
Application Method:	WaterlooWorks
Posting Goes Live:	05/13/2026 10:54 AM
Job Posting Status:	Approved

Company Information

Organization	University of Calgary
Division	McCaig Institute for Bone and Joint Health

Job Posting Information

Work Term	2026 - Fall
Job Type	Co-op Main
Job Title	Software Development Research Assistant
Number of Job Openings	1
Level	Junior Intermediate Senior
Region	Western Canada
Job - Address Line One	3280 Hospital Drive North West
Job - Address Line Two	HRIC Building
Job - City	Calgary
Job - Province/State	Alberta
Job - Postal/Zip Code	T2N 4Z6
Job - Country	Canada
Employment Location Arrangement	In-person

Additional Employment Arrangement Location Information

Work location is at the Foothills Medical Centre campus of the University of Calgary, in the Centre for Mobility and Joint Health.

Work Term Duration	4 month work term
---------------------------	-------------------

Job Summary

Our lab aims to better understand the development and progression of bone damage in arthritis. We use multi-modal imaging, including high resolution computed tomography (CT), 4DCT, magnetic resonance imaging (MRI) and ultrasound to quantify the changes that occur in an individual over time, and estimate joint loads. One feature that changes rapidly are erosions that occur in the joint, and can be observed with CT. Previous co-op students worked to develop modules in an open-source image processing that includes tools for image registration, segmentation and feature extraction/quantification such as erosions in multi-modal imaging in rheumatoid arthritis applications. We are actively expanding this toolkit to include additional modules and process existing data. Development will be done using 3D Slicer and Python (ITK, VTK) toolkits in the underlying framework.

The student will work in a research lab environment (<https://www.manskelab.ca/>) and have the opportunity to interact with graduate students and post-doctoral fellows, and take part in trainee activities within the McCaig Institute for Bone and Joint Health and Department of Radiology (including weekly seminar series, career development opportunities). The student may have the

opportunity to contribute to scientific publications (e.g., conference abstracts, manuscripts).

Job Responsibilities

- Experience with software development, preferably using Python.
- Familiarity with git, github and good documentation practices.
- Previous experience with image processing is an asset.
- Creativity and willingness to exchange ideas with others.
- Ability to work independently with little supervision, as well as in a team.
- Excellent communication and organizational skills.

Required Skills

- Experience with software development preferably using Python.
- Familiarity with git, github and good documentation practices.
- Previous experience with image processing is an asset.
- Creativity and willingness to exchange ideas with others.
- Ability to work independently with little supervision, as well as in a team.
- Excellent communication and organizational skills.

Compensation and Benefits

\$2500 - \$2850/month, for 35 hours of work per week.

Application Information

Application Delivery WaterlooWorks

Additional Application Information

In cover letter please specify:

- reason for interest in position
- goals of gaining experience in research